

MIDHUN PRAHASH SR

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Education

St. Joseph's Institute of Technology
B.Tech in Artificial Intelligence and Data Science

Expected 2027
CGPA: 8.4/10

Experience

Cospin

Jan 2026 – Mar 2026

AI Engineering Intern

- Built Python and FastAPI voice microservices for real-time speech ingestion, LLM orchestration, transcript persistence, and audio delivery in a conversational AI platform.
- Engineered streaming ASR and TTS services using Google Speech-to-Text v2, Google Cloud Text-to-Speech, WebSockets, containerization, and low-latency inference workflows for real-time AI interactions.

Lightcast

Sep 2025 – Jan 2026

AI/ML Intern

- Built a GraphRAG-based hybrid search and skill normalization pipeline to extract professional skills from unstructured job descriptions and map them to standardized taxonomy IDs.
- Designed and populated a Neo4j knowledge graph with aliases, category and subcategory relationships, and vector indexes to enable semantic retrieval over raw taxonomy data.
- Implemented a multi-stage RAG workflow using Gemini 2.5 Flash and all-mpnet-base-v2 embeddings to retrieve relevant context, generate structured JSON outputs, and reduce LLM hallucinations.
- Engineered a four-tier normalization system combining exact match, alias resolution, semantic similarity search, hierarchical contextual search, and threshold tuning to improve precision for ambiguous skill mapping.

Orion Governance

Jun 2025 – Jul 2025

Machine Learning Intern

- Developed a fuzzy string matching-integrated Random Forest classifier in Scikit-learn and Pandas, achieving 89% accuracy in identifying unauthorized employee usernames.

Projects

StudentAI | Python, FAISS, Sentence Transformers, OpenAI, Pytesseract, Prompt Engineering *RAG Learning Assistant*

- Built an end-to-end RAG pipeline for educational content with OCR ingestion, semantic chunking, embedding generation, FAISS vector indexing, dense retrieval, and context-grounded prompting.
- Integrated OpenAI LLM responses with retrieved evidence to generate structured What, Why, and How concept explanations aligned to source material and designed a retrieval-grounded question generation workflow.
- Improved answer quality by combining semantic retrieval, chunk overlap, grounding constraints, and prompt design to reduce unsupported responses.

UniGuide | FastAPI, PostgreSQL, Supabase, pgvector, JWT, RLS, Hybrid Retrieval *Production RAG System*

- Built a multi-tenant RAG backend with vector search, full-text search, BM25, reranking, semantic caching, and streaming responses for grounded answers over institutional knowledge bases.
- Designed 50+ SQL migrations, retrieval RPCs, vector indexes, chunk schemas, and org-scoped access controls to support scalable document intelligence workloads.

Username Matcher | Python, Scikit-learn, Pandas, Django, Fuzzy Matching *ML Security Application*

- Built and deployed a machine learning-based unauthorized username detection workflow using fuzzy string matching and a Random Forest classifier with 89% accuracy, then integrated it into a Django web application.

Review Analyzer | Selenium, NLTK, VADER, TextBlob, Matplotlib, Seaborn *Sentiment Analysis*

- Automated web scraping, sentiment classification, and visualization of product reviews using NLP libraries and data visualization workflows to summarize positive, negative, and neutral sentiment distributions.

Technical Skills

Languages: Python, Java, SQL

Machine Learning: Machine Learning, Classification, Supervised Learning, Data Modeling, Feature Engineering, Model Evaluation, Threshold Tuning, Statistical Modeling, Scikit-learn, PyTorch

NLP and LLM Systems: Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), GraphRAG, Information Retrieval, Semantic Search, Hybrid Search, Embeddings, Reranking, Prompt Engineering, LLM Orchestration, Model Serving, OpenAI API, Google Gemini, Sentence Transformers, LangChain, LlamaIndex

Data Systems and Infrastructure: FastAPI, PostgreSQL, Supabase, pgvector, FAISS, Neo4j, Knowledge Graphs, Vector Search, WebSockets, Server-Sent Events (SSE), Docker, Linux

Applied AI and Tooling: Automatic Speech Recognition (ASR), Text-to-Speech (TTS), Pandas, NumPy, NLTK, VADER, TextBlob, Selenium, Pytesseract, Matplotlib, Seaborn, Git, GitHub, GitLab

Publications

Midhun Prahash SR, Rhea Alphonsa Jose, Sam. V. George. "Enhancing Transformer-Based Hidden Suicidal Intention Detection with Sequential Modeling and Psycholinguistic Feature Fusion." In 2026 7th International Conference on Innovative Trends in Information Technology (ICITIIT - 2026), pp. 1-6. IEEE, 2026

Patent

Application No: 202541076262 - INTELLIGENT AGENT FOR PERSONALIZED & INCLUSIVE LEARNING SUPPORT SYSTEM FOR ADAPTIVE LEARNING WITH AGENTIC STRATEGIES

Achievements

Runner-up at the 2025 Thoothukudi District Police Cyber Hackathon (Top 2 of 800+ teams).